

Lecture 09-25

7.6 Weakly nonlinear oscillator | Huygens

$\epsilon \ll 1 \rightarrow$ weak/small nonlinear

$$\ddot{x} + x + \epsilon h(x, \dot{x})$$

Van der pol

$$+ \epsilon (x^2 - 1)$$

Duffing oscillator

$$+ \epsilon x^3$$

$\rightarrow$ See Bender & Orszag ch. 7 ex. 1

• roots of a cubic polynomial (not via the cubic formula)

• consider the expression

$$\epsilon = 0.001$$

$$x^3 - (4 + \epsilon)x + 2\epsilon = 0$$

• since ϵ is small, the roots are only slightly different from those of $x^3 - 4x = 0$. Consider a perturbation series $x = \sum_{n=0}^{\infty} a_n \epsilon^n$

$$x = \{ 2 + h_+(\epsilon), h_1(\epsilon), h_-(\epsilon) - 2 \}$$

$$x_- = 2 \text{ case: } h_-(\epsilon) - 2 \equiv x_-(\epsilon) = -2 + a_1 \epsilon + a_2 \epsilon^2 + a_3 \epsilon^3 + \dots$$

$$x_-^3(\epsilon) = (-2)^3 + 3(-2)^2(a_1 \epsilon + a_2 \epsilon^2 + a_3 \epsilon^3) + 3(-2)\epsilon^2(a_1 + \dots)^2 + \dots$$

$$\Rightarrow 0 = 0 + \underbrace{(12a_1 - 4a_2 + a_3)}_{\dots} \epsilon + (\dots) \epsilon^2 + \dots$$

$\rightarrow$ each coefficient must vanish individually

$$\epsilon: 8a_1 + 4 = 0$$

$$\Rightarrow a_1 = -\frac{1}{2}$$

$$\epsilon^2: a_2 = \frac{1}{8}, \text{ and so on}$$

"Now that this is ϕ^3 -field theory on a lattice..."

$$x_- = -2 - \frac{1}{2}\epsilon + \frac{1}{8}\epsilon^2 + \dots \quad (-2, 0, 2)$$

$$x_0 = 0 + \frac{1}{2}\epsilon - \frac{1}{8}\epsilon^2 + \dots$$

$$x_+ = 2 + 0\epsilon + 0\epsilon^2 + \dots \quad 0 \leftarrow \text{symmetry around } x=2!$$

$\Rightarrow$ Should have translated coords.

Regular Perturbation Theory

ϵ still small
 $\Rightarrow$ series terms decrease progressively

$$x(t) = x_0(t) + \epsilon x_1(t) + \epsilon^2 x_2(t) + \dots$$

$\rightarrow$ weakly linearly damped oscillator

$$\ddot{x} + x + 2\epsilon\dot{x} = 0 \quad (\text{ch 5!}) \quad x(t) = \frac{1}{\sqrt{1-\epsilon^2}} e^{-\epsilon t} \sin(t\sqrt{1-\epsilon^2})$$

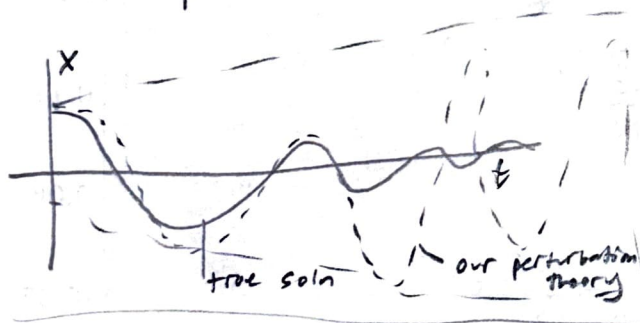
$$\epsilon^0: \ddot{x} + x_0 = 0 \quad x_0(t) = \sin t$$

$$\epsilon^1: \ddot{x} + x_0 + 2\epsilon\dot{x} = 0 \quad x_1(t) = -t \sin t$$

$$x(t) = \sin t - \epsilon t \sin t$$

$\rightarrow$ as ϵ grows above 1, system is overdamped and sin term becomes sinh (hyperbolic decay)

$\rightarrow$ for $\epsilon \ll 1$, it's underdamped and has a slow spiral



WKB Approximation

$\rightarrow$ atomic timescale t w/ slow, macroscopic timescale $T \gg t$

$\rightarrow$ We can represent this with a perturbation theory

$\rightarrow$ assume t, T are independent

$$x(t) = \dots e^{-\epsilon t} \sin(t + \dots)$$

$$T = \frac{1}{\epsilon} t \quad \epsilon \ll 1$$

$$\text{let } t \rightarrow \tau$$

$$x_c(t) = r(T) e^{\frac{i}{\epsilon} S_c(x)}$$

overall amplitude changes slowly

suprafast timescale of waves (atomic scale)

2 timescales

$$\text{let } \frac{d}{dt} = \frac{d}{d\tau} + \frac{d}{dT}$$

$\hookrightarrow$ secular expansion

$\rightarrow$ consider the "difference" from the zeroth order analytical solution (keplerian ellipse) to see a "longer" periodicity in the system

$\rightarrow$ systematically advance to longer and longer timescales of variations/periodicities in the system

$\rightarrow$ celestial dynamics on ~ 1 Gyr scales